An Efficient and Robust Random Forest Algorithm for Crop Disease Detection

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Abstract*- Nourishment security is a notable risk to crop diseases, however the testimonial stays unmanageable in various countries of the world due to the lack of proper foundation. This study is for productive crop organization in large areas using the key variables namely, texture, phenology, soil moisture, topographic vegetation, different satellite, and climatic data (precipitation and temperature). Since machine learning methodology in the sector of leaf-based image organization has displayed magnificent outcomes, an efficient Learning algorithm to find the impending disorder existing in plants on a massive scale, is used. In this system topographic and climate variables associated with spectral responses are compared and the near-infrared band is used with high spectral range (0.85 to 0.88m). The characteristic feature in image is obtained using Histogram of an Oriented Gradient (HOG). To evaluate RF models, a 20% independent dataset of training samples is used in addition to OOB data. The mean drop in accuracy and mean drop in Gini score are calculated. A comparative analysis is done on different Machine learning algorithms. The proposed RF model is efficient and robust algorithm obtaining an accuracy of 97.2% in detecting the disease to provide nourishment security.

Keywords-Machine Learning, Random Forest, HOG, Yield Prediction, Gini score.

I. INTRODUCTION

Birth to civilization is given by Agriculture. India is an agrarian country based upon crop production. Due to the rising population in countries like India, who have an ever-increasing need of food owing to the large population, agriculture stands as the backbone, and it is needed to fulfill the high-level needs. The Indian economy needs a huge up-gradation of the agriculture sector is mandatory to withstand the oscillating conditions. For flawless yield, a technology support is needed for periodic monitoring of crops should be healthy. Both quantity & quality of agricultural products are some of the factors which indirectly affect the significance. To increase the production & control disease, several pesticides are available. Though figuring out recent disease is time consuming and expensive, an effective and appropriate pesticide to curb the contaminated disorder is a challenging factor. The symptoms on leaves are mainly reflected by the presence of diseases on the plant. There is a need for automatic, accurate and less expensive techniques to address this challenge. The diseases affected by the wheat plants are stem and leaf rust. The Stem rust are small brown lesions on the leaves which occur initially and then blister like lesions develop on the leaf sheath. The disease extends from the leaf blade to the stem node. Leaves sheaths & White lesions on leaf cause powdery mildew. When the disease is severe, awns & glumes are also affected. Wiping away by abrasion of fingers over the infected regions of fungal escalation is hugely limited to exterior crop texture. Yellow rust is a disease where the crops color turns yellow, and the photosynthesis stops. All the disease which affects the productivity of the crops.

To escalate the accuracy rate & the recognition rate of the results, ML & DLA have been exercised. For crop diagnosis & disease detection, conventional ML approaches like SVM, CNN, random forest, ANN, fuzzy logic & K-means algorithm plays a dynamic role in the agriculture sector of research. The random forests learning method is used for classification and regression as it performs well on over-fitting of the training data sets. As a part of image & computer vision processing for object detection in diseased plants. The shape of the leaves is extracted by the HU moments. The leaf texture and color are extracted by haralick texture and the color descriptions in an image is done using histogram. The extracted features are used for the classification to predict the disease affected in the crops.

II. LITERATURE SURVEY

Machine learning plays a vital role for disease detection in plants where it helps the farmer to choose the pesticides for the affected plants. The authors have used classification algorithms for detection of diseased and healthy crops [1]. The authors [2] have used soybean dataset and detected three different kinds of diseases: Septoria leaf blight, Frog eye, & downy mildew in the diseased leaves. The system has produced an average classification accuracy of approximately 90%. The authors [3]

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have used grape leaves for detection of esca and blight and produce a classification accuracy of 91%. The authors have used LAB and HSI color models to extract features.

The authors [4] have used palm leaf dataset for detection of diseases like Anthracnose & Chimera**.** The system targets to get around 95% accuracy respectively. The authors [5] created a system to increase the precision in distinguishing 2 classes of illness triggered by fungi like Powdery Mildew & Downy Mildew in cucumber plants. The authors introduced a system using back-propagation algorithms to classify & recognize disorders of pomegranate plant & the experimental result of the system gives 90% correctness.

ANN is another computational model at pattern recognition & ML which have been used by most of the authors for the identification and detection of disease in plants. The system detected the disease like tiny whiteness, late scorch, & cottony mold on the surface of the leaves. The authors [6] have done research over a series of ML algorithms & highlighted the importance of each algorithm. The analysis is done on the tomato plants for identification and detection of the disease.

The authors [7] have made use of SVM Classifiers in detection of disease in varied citrus plants like oranges, lemons & grapefruit leaf. They are affected by anthracnose & canker disease. The experimental result obtained by the classifier is 95%. The authors have used Grape plant for detection of downy mildew disease and produces an accuracy of 88.89%. The authors [8] have researched potato plants for detection of Early blight & blight disease around three hundred pictures that are available and produce around 95% accuracy.

The authors [9] have used a tea plant dataset to detect disease using an SVM classifier and have produced 90% accuracy. They have done analysis for "rice disease detection using deep learning". They have proposed an automated disorder analyzing technique for detecting disease of paddy. The metadata was practiced with the help of 3 CNN & attained a high precision of around 99.53%. The authors have done comprehensive analysis on different machine learning algorithms to analyze plant disorder. The categorization techniques used are ANN, CNN, SVM, K-nearest neighbor classification technique & fuzzy classification methods from which Convolutional Neural Network produces high accuracy.

From the survey of the related works all the authors have used various ML algorithms to do the analysis on the crop for disease identification and detection of diseases affected in the plants for producing high yield. The results obtained produces a quantitative and qualitative production. The technological development in the farming helps the farmers with a digitized solution. This leads to avoid time consuming process and avoid excess use of pesticides to the crops.

III. METHODOLOGY

The proposed work is shown in figure 1 which shows the methodology of early identification and detection of the disease affected by the crops. Raw data collected is pre-processed then sampled. The data is collected from the government website and then the sampled data is then feature extracted. The extracted features are then used for the classification. A comparative analysis is done on various learning algorithms. Finally, the model is evaluated using performance metrics for finding the accuracy of the model. The model produces efficient and effective results.

A. Logistic Regression:

It's one among statistical models in logistic function used for base form binary dependence. Most of the researchers have used rice, wheat, barley, tomato, and chilly datasets. Logistic Regression is used for the identification of disease in these plants and detecting the area where it is affected. The regression model produces good results.

B. Multinomial Naïve-Bayes

In this multinomial event model, attributes are displayed in airwaves of facts and the feature is portrayed by a vector x= (x1, x2, x3,…,xn). This will be deliberated as a histogram where xi = frequency of event. As the Bernoulli Naïve-Bayes, the same is used in classification of the diseased and healthy plants.

C. Random Forest Classifier

This technique uses construction of decision trees for the **classification** with tuples from data meant for model training.

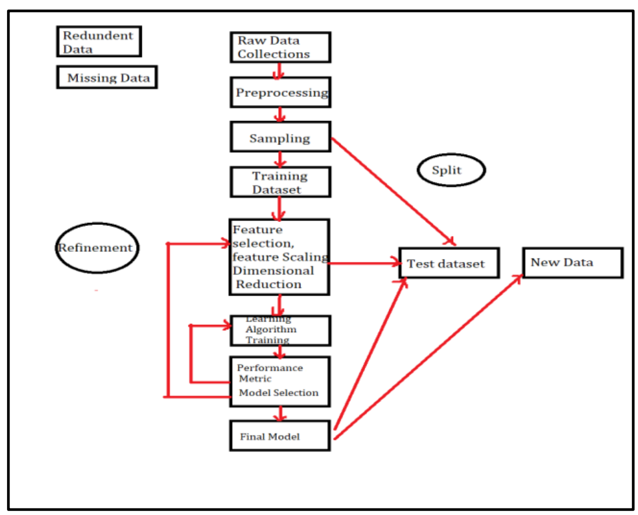


Figure 1: Leaf disease detection using Random Forest

Every internal node/non-leaf describes a tree structure constraint to check in the leaf nodes & attributes. The idea in this type is that the knowledge of groups. This comprises an outsized no. of decision trees in which every individual tree is employed in the forecast of a category. This depends on multiple decision trees instead of one decision tree. Accuracy of this sort of classifiers are often up to 90%.

D. Support Vector Machine

It's one among strong supervised learning methods for data classification and regression tasks. SVM can divide the crop and find the exit crop for present season conditions. This classification model ensures the accuracy using the hyperplanes and this reduces the structural risks.

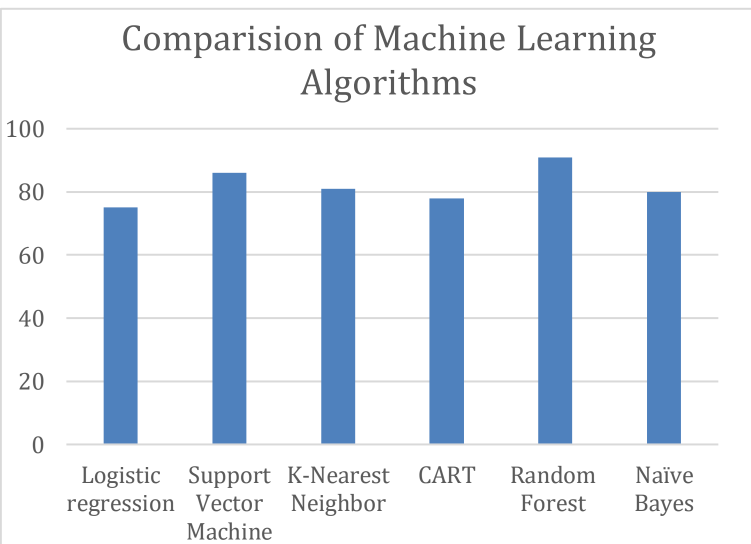
A. Convolutional Neural Network

CNN which comprises different tiers like input tiers, output tiers alongside some hidden tiers which include pooling tiers, convolutional tiers, RELU tiers, fully connected tiers, normalization tiers. This is often a deep learning algorithm that also gives positive results on the disease metadata of the crops.

IV. PERFORMANCE ANALYSIS AND RESULTS

A. Identify and proper missing data

At frequent intervals we get to understand that the information



collected cannot be available for 24X7 days and there might certainly be some minor or major missing parameters within the obtained data. Procedure to see the correctness of the obtained data with a number of information missing by drawing basic plots. Our preference is towards spotting anomalies during a graph instead of in numbers. So, by this manner we will overcome the issues.

B. Prepare the info for model

We need to make many minor changes to convert our data within the machine-understandable format. In our paper we've made use of the Python for implementing the frames on the structure which may be called as a data frame, which may be a table with columns and rows. The essential steps for conversion of the info will depend upon the type of algorithm used for the model and the metadata accumulated, though few data are often manipulated.

C. Establish a baseline model

We need to establish a baseline, before we could evaluate and make predictions, an ideal criterion that we beat with the baseline model. While at any time if our model missed to enhance upon the baseline, then it might turn out as a loss and that we need to try another algorithm or conclude that our problem isn't suitable for machine learning. The prediction in this case is that the season, climate, humidity, spread of pest and wind speed averages.

D. Train the model on the training data

Using Scikit-learn, all the methods of knowledge creation, preparation and training the system is basically easy. we've imported the RFR model using scikit-learn, instantiated the system, & met the system on the schooling data.

E. Make predictions on the test data

At this moment our system is now schooled to find out the interconnection among all targets and therefore the features. The very post step after this is often to form out how perfect our system works. After comparisons will be made with all the estimations to the known outputs. While doing regression, we need to confirm the form use of absolutely the flaws as we anticipate many of our outputs to be less & a few values are more. The prediction shows that random forest algorithm performs well on the dataset.

F. Performance Metrics

To put our ideas in point of view, accuracy is often calculated with the help of mean average % error different from 100% error.

Figure 2: Performance analysis of algorithms

In Figure 2, we have compared all existing machine learning algorithms. And we could conclude that k-nearest neighbor algorithm and Random Forest algorithm are the best algorithms in comparison.

This study helps us to search the simplest ML algorithm for the suggested system Random Forest algorithm & CNN stipulates supreme processing speed & accuracy equated to another model.

V. CONCLUSION

This deep driven study highlighted that the special publications make use of a change of characteristics, based on the confinement of the study and the accessibility of metadata. Each paper examines prediction with ML but differs from its features and this may as well vary in crop, geological scale & position. The aim of the research and the availability of the dataset is dependent on the choice of these features. For the yield prediction, the models with high characteristics always did not give the fullest performance in education. To figure out the best analyzing model, systems with additional one or two features should be tested. Different Studies have been used in many algorithms. The result clearly shows that a set of specific machine learning algorithms are used repetitively and still it's very hard to conclude on which is the best algorithm in those sets of algorithms. Among the models, some models like gradient boosting tree, neural network & random forest algorithm were quite famous. To analyze which could be the best model for prediction, most of the studies used several types of ML models. For crop yield prediction, the most applied & the most appealing algorithm is neural networks. Since the proposed well focused to research deeper on what is the width of deep learning. To conclude we believe that the Random Forest Algorithm is best in comparison with all the other algorithms.

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